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In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

In [2]: # Section C - Mini Capstone Project
# Food Delivery Analytics Console - Full EDA Repor

np.random.seed(30)

#Generating dataset
restaurants = ["Spice Hub", "Urban Bites", "Tasty Corner", "Food Street", "Royal Kitchen"]
cities = ["Vadodara", "Ahmedabad", "Surat"]
cuisines = ["Indian", "Chinese", "Fast Food"]

df = pd.DataFrame({"restaurant_name": np.random.choice(restaurants, 250),"city": np.random.choice(cities, 250),
"order_value": np.random.uniform(150, 1000, 250),"delivery_time_mins": np.random.normal(30, 7, 250),
"rating": np.random.uniform(1.0, 5.0, 250),"cuisine_type": np.random.choice(cuisines, 250)})
```

```
In [3]: df
```

Out[3]:

	restaurant_name	city	order_value	delivery_time_mins	rating	cuisine_type
0	Royal Kitchen	Vadodara	744.871000	45.258617	2.312274	Indian
1	Royal Kitchen	Surat	807.332157	33.428260	3.644407	Chinese
2	Tasty Corner	Vadodara	474.007802	21.751405	2.552070	Chinese
3	Food Street	Surat	176.547952	28.172788	3.926836	Indian
4	Urban Bites	Ahmedabad	693.793930	35.976107	3.502453	Fast Food
...	...	...	...	...	...	...
245	Tasty Corner	Vadodara	860.039367	38.158094	2.394128	Fast Food
246	Tasty Corner	Vadodara	823.623076	27.472385	1.020470	Indian
247	Urban Bites	Vadodara	518.487789	17.651599	1.243640	Indian
248	Urban Bites	Ahmedabad	566.354380	31.607754	1.440648	Fast Food
249	Tasty Corner	Ahmedabad	190.826481	43.635083	3.785232	Indian

250 rows × 6 columns

```
In [5]: # Creating a few missing numeric values for the required null-handling step

for column in ["order_value", "delivery_time_mins", "rating"]:
    missing_rows = np.random.choice(df.index, size=5, replace=False)
    df.loc[missing_rows, column] = np.nan

numeric_columns = df.select_dtypes(include=np.number).columns
for column in numeric_columns:
    df[column] = df[column].fillna(df[column].median())
```

```
In [15]: def summary_statistics():
    print("\n--- Summary Statistics ---")
    print(f"Mean order value: Rs {np.mean(df['order_value']):.2f}")
    print(f"Mean delivery time: {np.mean(df['delivery_time_mins']):.2f} minutes")
    print(f"Mean rating: {np.mean(df['rating']):.2f}")
    print(f"Delivery time standard deviation: {np.std(df['delivery_time_mins']):.2f}")
```

```
In [25]: def distribution_analysis():plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.hist(df["order_value"], bins=15)
plt.title("Order Value Distribution")
plt.xlabel("Order Value")
plt.ylabel("Frequency")
plt.subplot(1, 2, 2)
plt.hist(df["delivery_time_mins"], bins=15)
plt.title("Delivery Time Distribution")
plt.xlabel("Delivery Time")
plt.ylabel("Frequency")
plt.tight_layout()
```

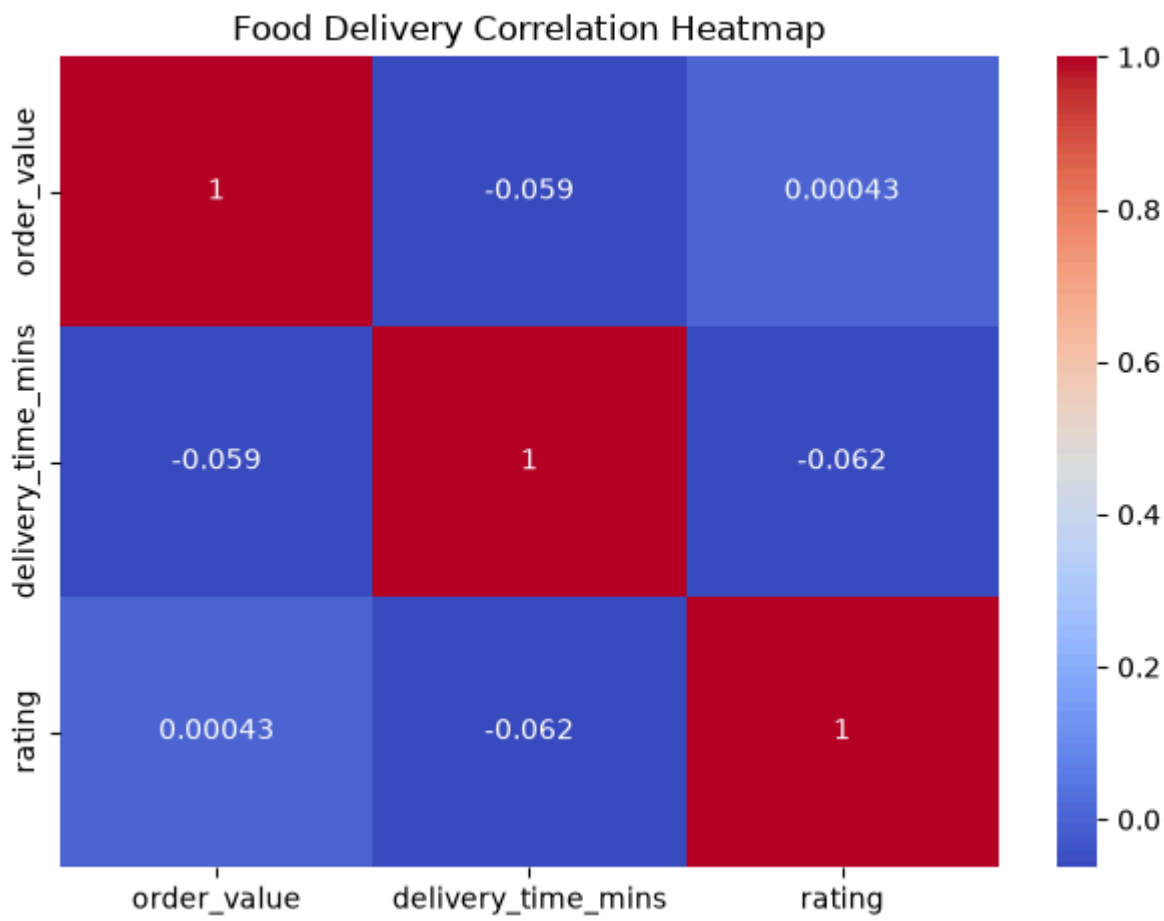


```
In [27]: plt.show()
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In [28]: plt.savefig("distribution_analysis.png")
print("Graph saved as distribution_analysis.png")
```

Graph saved as distribution_analysis.png
<Figure size 640x480 with 0 Axes>

```
In [31]: def correlation_heatmap():plt.figure(figsize=(8, 6))
sns.heatmap(df.select_dtypes(include=np.number).corr(),annot=True,cmap="coolwarm")
plt.title("Food Delivery Correlation Heatmap")
plt.tight_layout()
plt.savefig("correlation_heatmap_capstone.png")
plt.show()
```



```
In [33]: def restaurant_performance():  
  
    result = df.groupby("restaurant_name")[["order_value", "delivery_time_mins", "rating"]].mean()  
  
    print("\n--- Restaurant Performance Report ---")  
    print(result)
```

```
In [ ]: def final_report():  
  
    performance = (  
        df.groupby("restaurant_name")["rating"]  
        .mean()  
        .sort_values(ascending=False)  
        .head(3)  
    )  
  
    corr = df.select_dtypes(include=np.number).corr().abs()  
    corr = corr.where(corr != 1)  
  
    strongest_pair = corr.stack().idxmax()  
    strongest_value = corr.stack().max()  
  
    mean_delivery = df["delivery_time_mins"].mean()  
    std_delivery = df["delivery_time_mins"].std()  
  
    print("\n===== FINAL REPORT =====")  
  
    print("\nTop 3 restaurants:")  
    for i, (restaurant, rating) in enumerate(performance.items(), 1):  
        print(f"{i}. {restaurant}: {rating:.2f}")  
  
    print("\nStrongest correlation:")  
    print(f"{strongest_pair[0]} vs {strongest_pair[1]}: {strongest_value:.2f}")  
  
    print(f"\nAverage delivery time: {mean_delivery:.2f} minutes")  
    print(f"Delivery time standard deviation: {std_delivery:.2f} minutes")  
  
while True:  
  
    print("\n===== FOOD DELIVERY ANALYTICS CONSOLE =====")  
    print("1. Summary Statistics")  
    print("2. Distribution Analysis")  
    print("3. Correlation Heatmap")  
    print("4. Restaurant Performance Report")  
    print("5. Exit")  
  
    choice = input("Enter your choice: ")  
  
    if choice == "1":  
        summary_statistics()  
  
    elif choice == "2":  
        distribution_analysis()  
  
    elif choice == "3":  
        correlation_heatmap()  
  
    elif choice == "4":  
        restaurant_performance()  
  
    elif choice == "5":  
        final_report()  
        print("\nProgram ended.")  
        break  
  
    else:  
        print("Please enter a valid choice.")
```

```
===== FOOD DELIVERY ANALYTICS CONSOLE =====  
1. Summary Statistics  
2. Distribution Analysis  
3. Correlation Heatmap  
4. Restaurant Performance Report  
5. Exit  
--- Summary Statistics ---  
Mean order value: Rs 567.31  
Mean delivery time: 29.86 minutes  
Mean rating: 2.88  
Delivery time standard deviation: 7.34
```

```
===== FOOD DELIVERY ANALYTICS CONSOLE =====  
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```

```
--- Restaurant Performance Report ---
      order_value  delivery_time_mins    rating
restaurant_name
Food Street      587.577228           28.853148  2.904346
Royal Kitchen    602.590788           30.198464  3.018801
Spice Hub        557.034588           28.478380  2.780605
Tasty Corner     594.014682           30.739303  2.720734
Urban Bites      496.599496           30.910050  2.944741
```

```
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In []: